TENSEGRITY PROJECT

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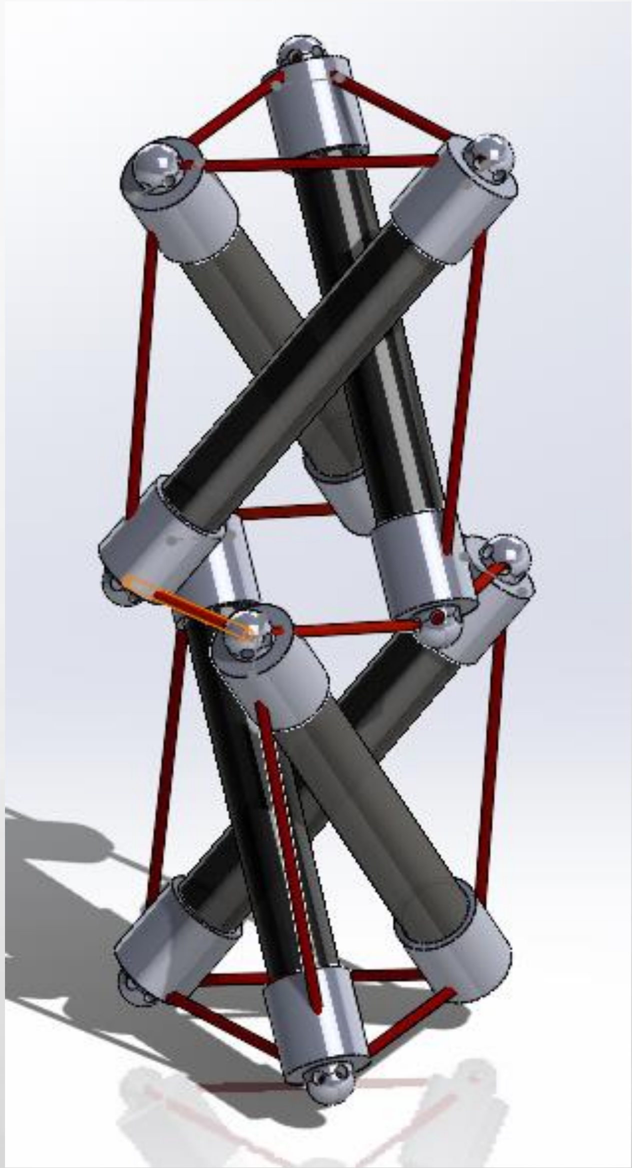
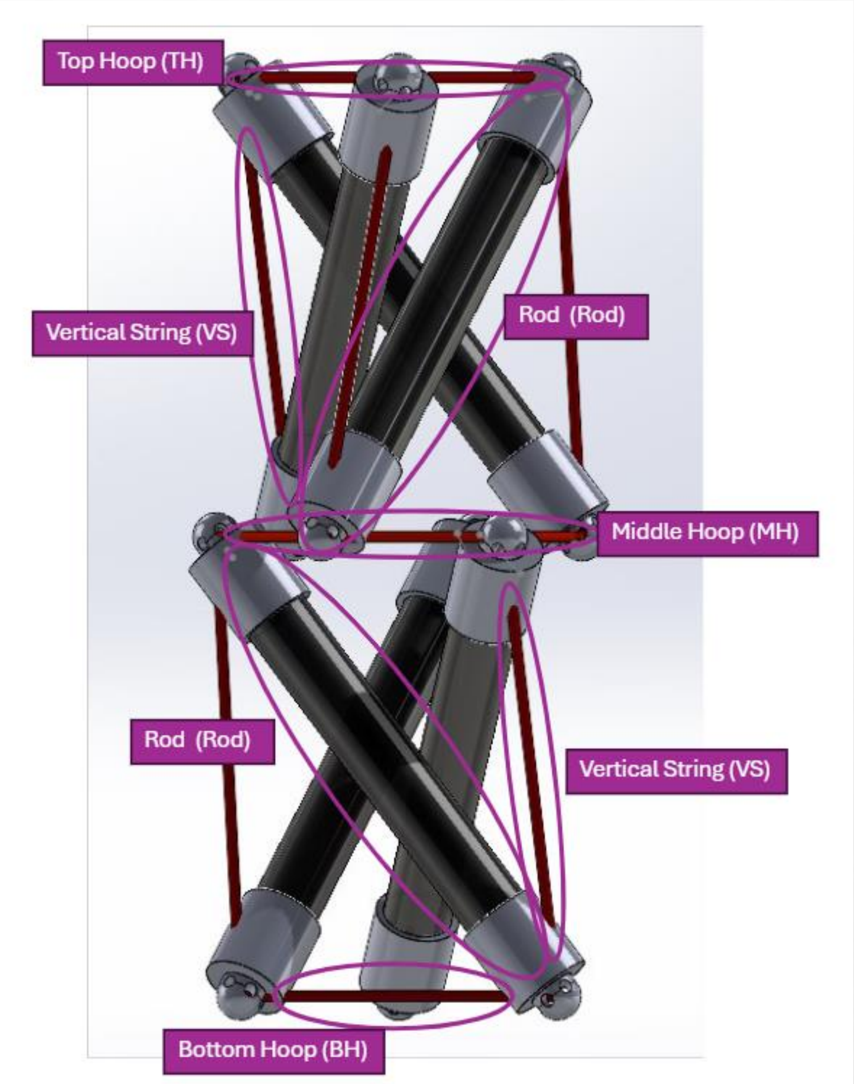
OVERALL CONSTRUCTION

- ACRYLIC RODS ARE IN A TRI-CONFIGURATION STACKED ON TOP OF EACH OTHER (X2)
- STRINGS OF VARYING LENGTHS ARE CONNECTED TO 3D PRINTED CAPS SUPPORTING TENSEGRITY STRUCTURE



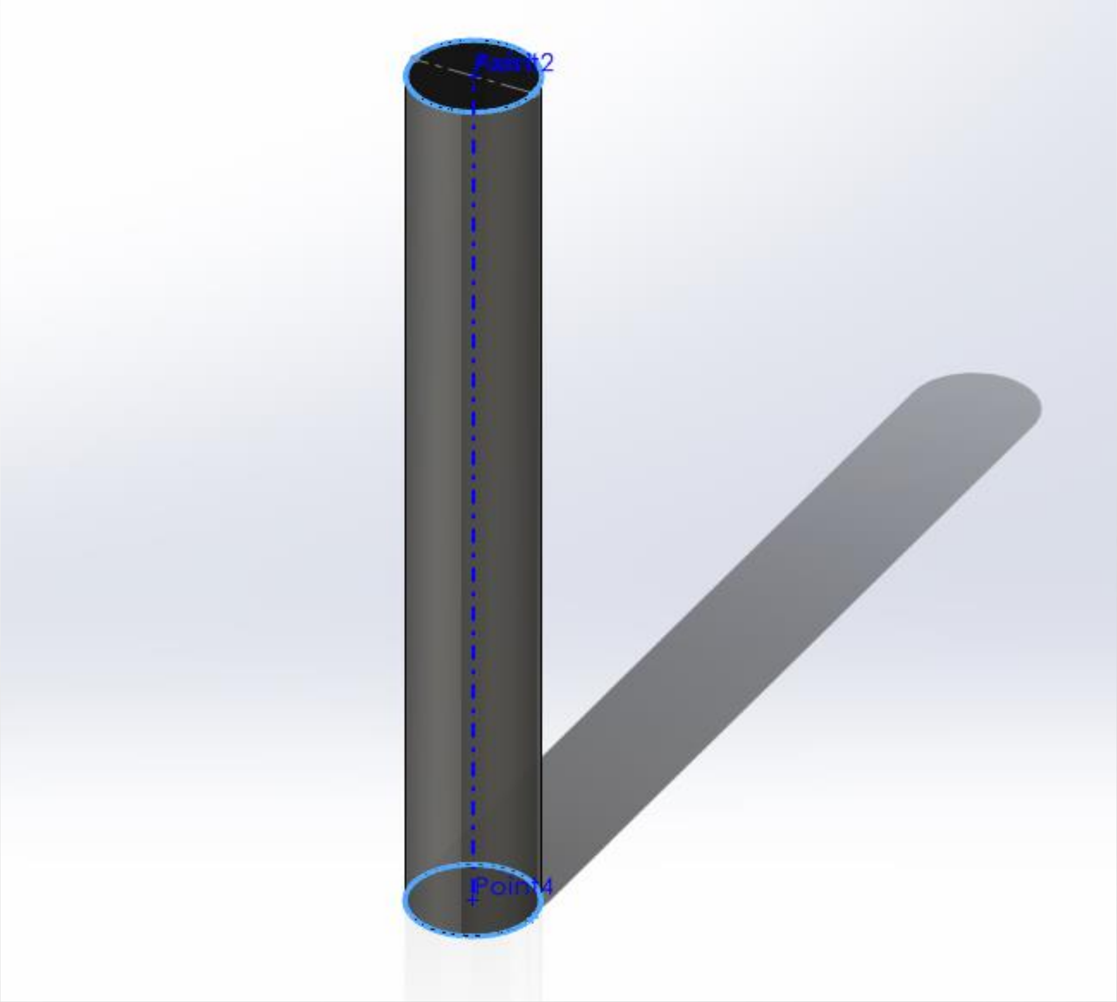
COMPONENTS



Top Hoop (TH)
Vertical String (VS)
Rod (Rod)
Middle Hoop (MH)
Rod (Rod)
Vertical String (VS)
Bottom Hoop (BH)

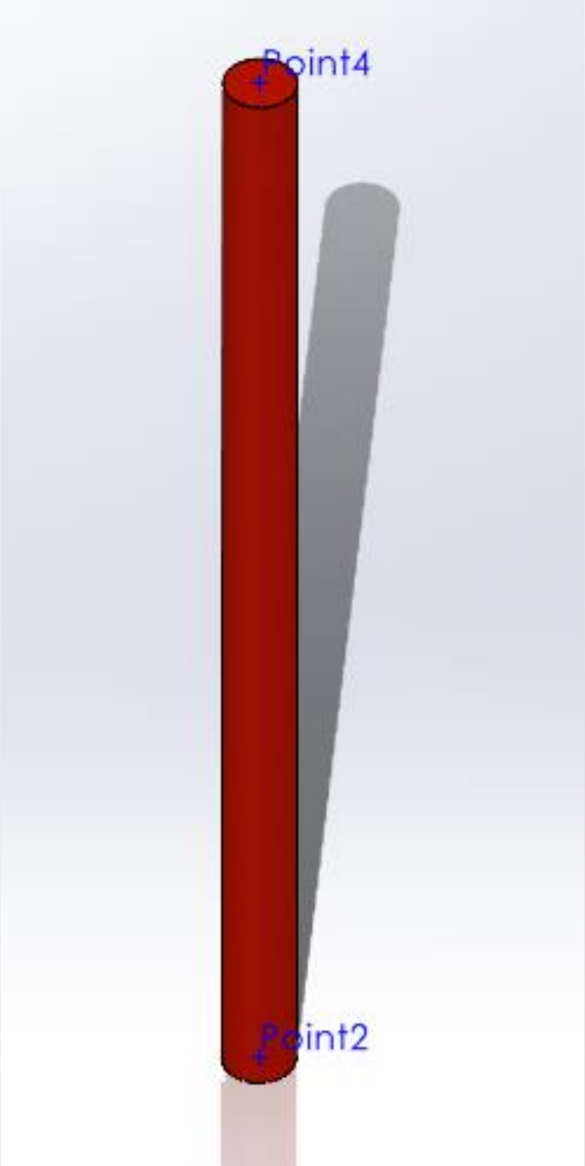
ACRYLIC ROD

126mm length
13.8mm diameter



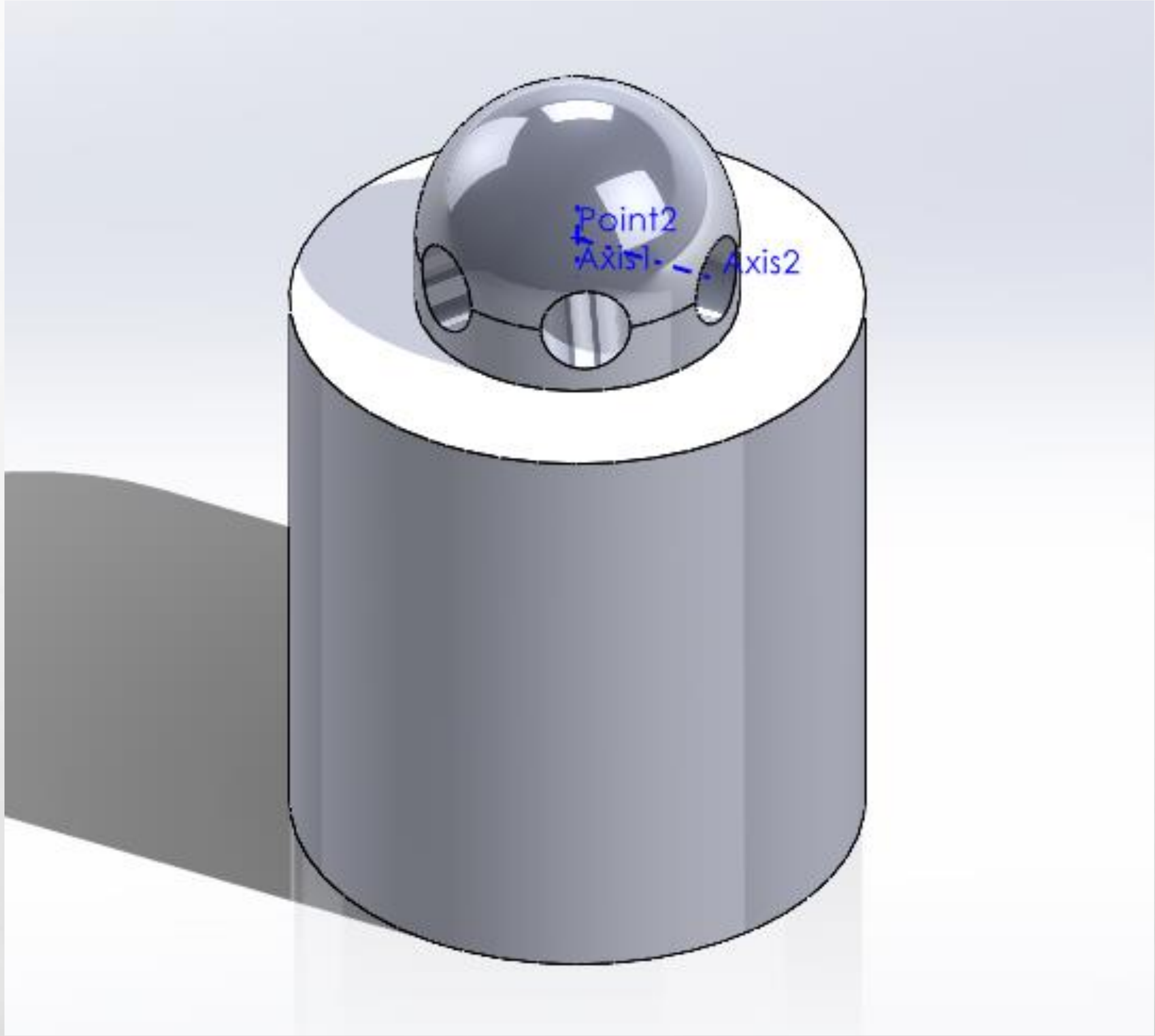
FISHING LINE (30LB MONO)

LINES WITH VARYING LENGTH ACCORDING TO LOCATION



END CAP

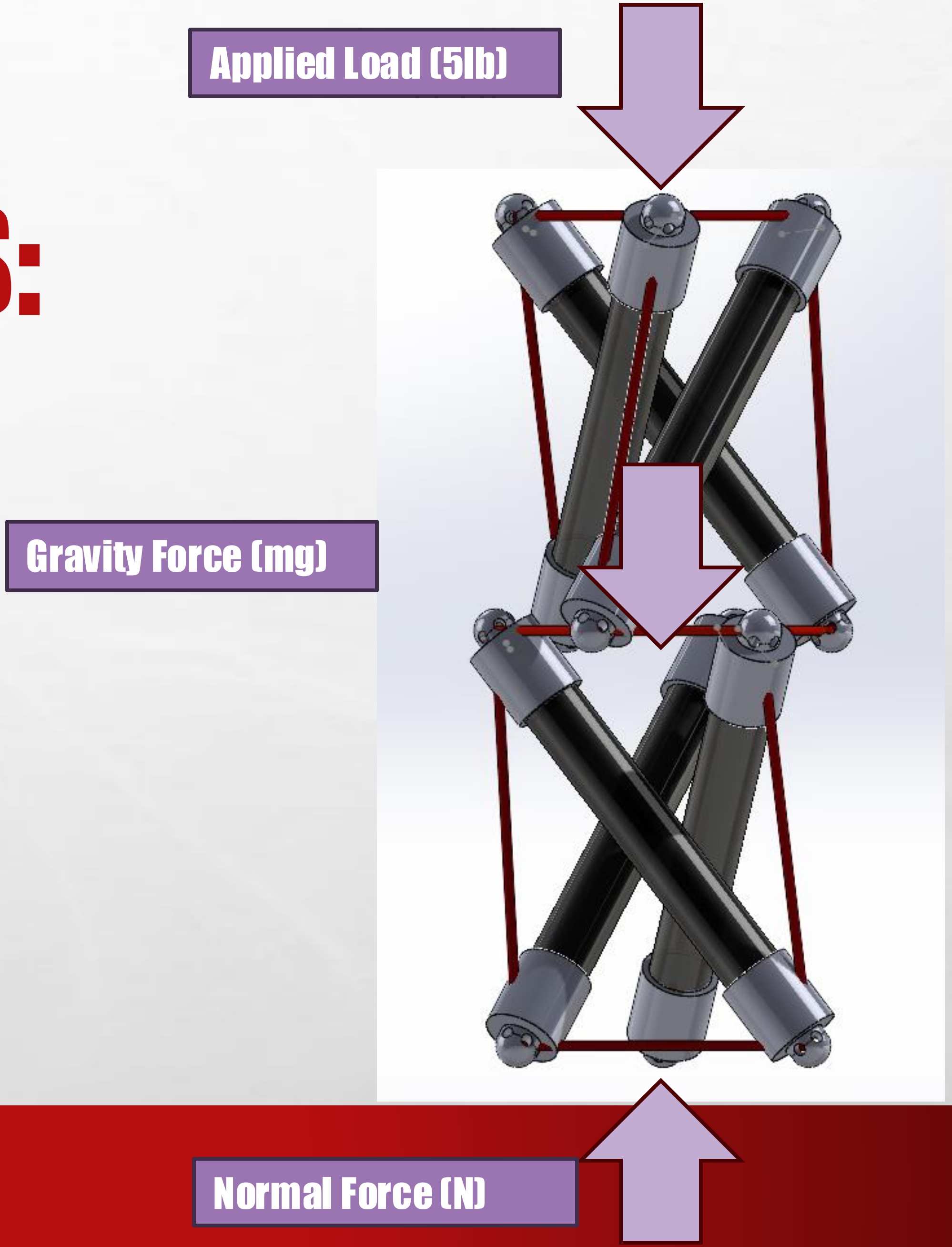
3D to fit snugly around the 13.8mm diameter acrylic rod



EQUILIBRIUM FORCES:

$$\sum \vec{F} = -F - mg + N = 0$$

$$N = F + mg$$

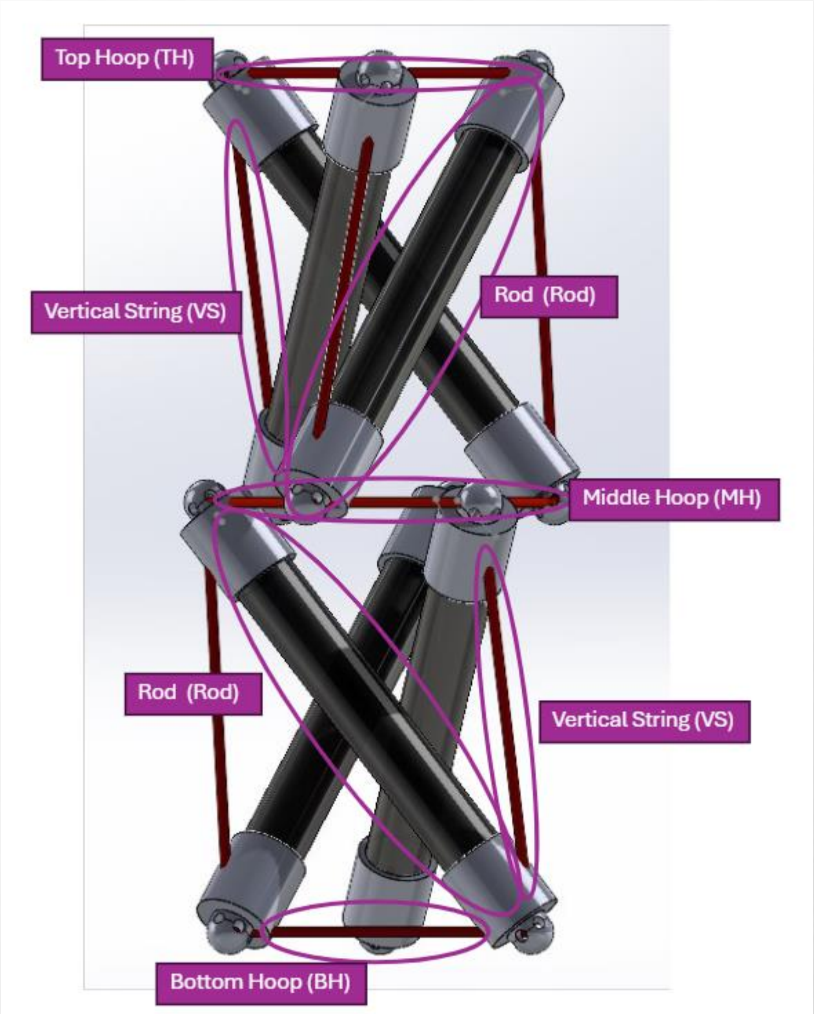


Top and Bottom Ends

Force	Direction	Magnitude
Compression	-0.5835**i**, -0.18397**j**, 0.7910**k**	F_{Rod}
Tension	0.8660**i**, 0.5**j**, 0**k**	$T_{BH\ or\ TH}$
Tension	0.8660**i**, -0.5**j**, 0**k**	$T_{BH\ or\ TH}$
Tension	0.03742**i**, 0.1687**j**, 0.9849**k**	T_{VS}
Normal Force or Load	0**i**, 0**j**, 1**k**	F_N

Middle Ring

Force	Direction	Magnitude
Compression	0.5835**i**, 0.18397**j**, 0.7910**k**	F_{Rod}
Tension	0.08715**i**, 0.99619**j**, 0**k**	T_{MH}
Tension	-0.9063**i**, 0.4226**j**, 0**k**	T_{MH}
Tension	-0.1275**i**, 0.1168**j**, -0.9849**k**	T_{VS}
Normal Force or Load	0**i**, 0**j**, -1**k**	F_{MHN}



EOM

$$\overrightarrow{F_{BH1}} = T_{BH}(0.8660i + 0.5j + 0k)$$

$$\overrightarrow{F_{BH2}} = T_{BH}(0.8660i - 0.5j + 0k)$$

$$\overrightarrow{F_{VS1}} = T_{VS}(0.03742i + 0.1687j + 0.9849k)$$

$$\overrightarrow{F_{VS2}} = T_{VS}(-0.1275 + 0.1168j - 0.9849k)$$

$$\overrightarrow{F_{MH1}} = T_{MH}(0.08715i - 0.99619j + 0k)$$

$$\overrightarrow{F_{MH2}} = T_{MH}(-0.9063i + 0.4226j + 0k)$$

$$\overrightarrow{F_{MHN}} = \frac{1}{3}\left(F + \frac{1}{2}mg\right)(0i + 0j - 1k)$$

$$\overrightarrow{F_g} = \frac{1}{6}mg(0i + 0j - 1k)$$

$$\overrightarrow{F_N} = \frac{1}{3}(F + mg)(0i + 0j + 1k)$$

$$\sum F_x = 0.866T_{BH} + 0.866T_{BH} + 0.03742T_{VS} - 0.1275T_{VS} + 0.08715T_{MH} - 0.9063T_{MH}$$

$$1.732T_{BH} - 0.0901T_{VS} - 0.8192T_{MH} = 0$$

$$\sum F_y = 0.5T_{BH} - 0.5T_{BH} + 0.1687T_{VS} + 0.1168T_{VS} - 0.9962T_{MH} + 0.4226T_{MH}$$

$$0.2855T_{VS} - 0.5736T_{MH} = 0$$

$$\sum F_z = 0.9849T_{VS} - 0.9849T_{VS} - \frac{1}{3}F - \frac{1}{6}mg - \frac{1}{6}mg + \frac{1}{3}F + \frac{1}{3}mg$$

$$0 = 0$$

EOM

$$\overrightarrow{M_N} = \overrightarrow{R_A} \times \overrightarrow{F_N} = 0$$

$$\overrightarrow{M_{BH1}} = \overrightarrow{R_A} \times \overrightarrow{F_{BH1}} = 0$$

$$\overrightarrow{M_{BH2}} = \overrightarrow{R_A} \times \overrightarrow{F_{BH2}} = 0$$

$$\overrightarrow{M_{VS1}} = \overrightarrow{R_A} \times \overrightarrow{F_{VS1}} = 0$$

$$\overrightarrow{M_{VS2}} = \overrightarrow{R_B} \times \overrightarrow{F_{VS2}} = (L \cdot T_{VS})(-0.2736i + 0.4738j + 0.09161k)$$

$$\overrightarrow{M_{MH1}} = \overrightarrow{R_B} \times \overrightarrow{F_{MH1}} = (L \cdot T_{MH})(0.7880i + 0.06894k - 0.5973k)$$

$$\overrightarrow{M_{MH2}} = \overrightarrow{R_B} \times \overrightarrow{F_{MH2}} = (L \cdot T_{MH})(-0.3343i - 0.7169j + 0.4133k)$$

$$\overrightarrow{M_{MHN}} = \overrightarrow{R_B} \times \overrightarrow{F_{MHN}}$$

$$= \left(L \cdot \left(\frac{1}{3}F + \frac{1}{6}mg\right)\right)(-0.1840i + 0.5835j + 0k)$$

$$\overrightarrow{M_g} = \left(\frac{1}{2}\overrightarrow{R_B}\right) \times \overrightarrow{F_g}$$

$$= \left(\frac{1}{12}L \cdot mg\right)(-0.1840i + 0.5835j + 0k)$$

$$\sum M_x = -0.2736LT_{VS} + 0.7880LT_{MH} - 0.3343LT_{MH} - 0.0613LF - 0.0307Lmg - 0.0153Lmg$$

$$-0.2736T_{VS} + 0.4537T_{MH} - 0.0613F - 0.046mg = 0$$

$$\sum M_y = 0.4738LT_{VS} + 0.06894LT_{MH} - 0.7169LT_{MH} + 0.1945LF + 0.0973Lmg + 0.04863Lmg$$

$$0.4738T_{VS} - 0.6480T_{MH} + 0.1945F + 0.1459mg = 0$$

$$\sum M_z = 0.09161T_{VS} - 0.5973T_{MH} + 0.4133T_{MH}$$

$$0.09161T_{VS} - 0.184T_{MH} = 0$$

EOM RESULTS

Total Structure Weight = mg

Total Applied Force = F

Tension in Top Hoop = 0.370F + 0.093mg

Tension in Top Vertical Strings = 1.287F + 0.322mg

Tension in Middle Hoop = 0.641F + 0.481mg

Tension in Bottom Vertical Strings = 1.287F + 0.966mg

Tension in Bottom Hoop = 0.370F + 0.278mg

Total Structure Weight ~ 5N

Total Applied Force ~ 25N

Tension in Top Hoop = 9.715N

Tension in Top Vertical Strings = 33.785N

Tension in Middle Hoop = 18.43N

Tension in Bottom Vertical Strings = 37.005N

Tension in Bottom Hoop = 10.640N

MATERIALS LIST:

Material	Cost
Acrylic Rods	Given
Fishing Line	Given
End Caps	3-D Printing (Free)
Glue	Have on Hand